

RESILIENT BUT NOT RESTORABLE?

An Empirical Test of the Resilience–Restorability Distinction in Canadian Healthcare

SignalRupture (SR)

Structural Restoration Dynamics Comparative Test

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Abstract

Structural Restoration Dynamics (SRD) survives its first challenge against resilience engineering only if restorability is empirically distinguishable from resilience. This paper directly tests that surviving claim. Rather than comparing unrelated sectors, it uses a within-system design based on Canadian healthcare, allowing multiple service trajectories to be examined under the same national institutional environment and largely comparable pandemic/post-pandemic period. Resilience is defined as the capacity to maintain or adapt service under disturbance; restorability is defined as the capacity to return toward a declared functional baseline while rebuilding durable slack and reducing extraordinary dependence. Four candidate trajectories are examined. Radiation therapy exhibited high resilience during the early pandemic and remained near benchmark performance, providing a High Resilience–High Restorability case. Cataract surgery suffered a major benchmark deterioration in 2020 but returned to approximately pre-pandemic performance by 2023–2024, providing a Low Resilience–High Restorability case. Hip and knee replacement services suffered pandemic disruption and remained below 2019 benchmark attainment in 2024 despite volumes substantially above 2019, providing a Low Resilience–Low/Incomplete Restorability case. Hospital-wide staffing and access provide a candidate High Resilience–Low/Incomplete Restorability condition: hospitals remained operational through large increases in overtime and purchased staffing while emergency-department and other access pressures persisted. The matrix therefore does not collapse into a single resilience axis. However, the strongest SRD claim remains provisional because resilience is multidimensional and the present operationalization is outcome-based rather than a complete measurement of resilience-engineering capacities. The paper specifies a falsification test: if independent

resilience measures fully predict restoration trajectories and the restorability dimension adds no prospective information, the distinction should be absorbed into resilience theory rather than maintained as an independent SRD construct.

Keywords: resilience; restorability; Structural Restoration Dynamics; Canadian healthcare; wait times; functional recovery; compensated continuity; SignalRupture

1. The Remaining SRD Claim

The prior adversarial comparison with resilience engineering removed several overly strong SRD assumptions. Elevated compensation is not inherently erosive. Resilience and compensated continuity are not mutually exclusive. Recovery objectives, restoration time, system capacity, and organizational action are not SRD inventions. After those concessions, one claim remained:

Resilience $\neq$ Restorability

This paper attempts to break that claim.

Resilience is operationalized here as the demonstrated capacity of a service or system to maintain or adapt function during substantial disturbance. Restorability is operationalized as the demonstrated capacity to return toward a declared functional baseline after impairment while reducing rather than indefinitely increasing extraordinary compensation.

Resilience_t = capacity to sustain/adapt function under disturbance

Restorability_t→_{t+k} = capacity to reduce restoration gap and rebuild durable operating margin

ADVERSARIAL TEST — Construct-overlap risk

Many resilience frameworks already include recovery and adaptation over time. Therefore the existence of a statistical or descriptive distinction is not enough. SRD must show that restorability can vary independently of observed resilience and eventually that the additional dimension improves prediction or intervention.

2. Why a Within-System Test

A cross-domain matrix risks confusing resilience/restorability with sector differences. Electricity, housing, hospitals, and legacy IT have different units, lead times, governance, and failure modes. Canadian healthcare offers a stronger first test because CIHI reports multiple services across a common health-system environment, including cataract surgery, joint replacement, radiation therapy, diagnostic imaging, emergency-department access, and hospital staffing.

The pandemic provides a common disturbance period, while 2019 serves as a practical historical comparison point for several indicators. This does not make 2019 an ideal normative target; it simply provides a declared baseline for the test.

Baseline class used here: B_H = 2019 historical performance

ADVERSARIAL TEST — Baseline criticism

A historical baseline can encode pre-existing dysfunction. Returning to 2019 therefore demonstrates restoration toward a prior state, not proof of optimal healthcare performance. The paper uses 2019 only to test separability of resilience and restorability.

3. The 2×2 Matrix

	High Restorability	Low / Incomplete Restorability
High Resilience	Radiation therapy: performance remained comparatively stable and near benchmark through disruption	Hospital-wide continuity: services maintained, but extraordinary staffing and access pressures persisted
Low Resilience	Cataract surgery: sharp 2020 deterioration followed by return to pre-pandemic benchmark performance	Hip/knee replacement: substantial disruption followed by incomplete benchmark restoration despite higher volumes

The matrix is provisional. 'High' and 'low' are comparative classifications derived from observed trajectories, not universal thresholds. The purpose is to determine whether all services line up along a single diagonal. They do not.

4. Quadrant I — High Resilience, High Restorability: Radiation Therapy

CIHI reported that wait times for radiation therapy remained relatively stable during the early pandemic even while many hospital services were disrupted. Its 2021 hospital-services analysis stated that wait times for radiation therapy remained stable through the first two waves. By 2023 and 2024, approximately 94% of patients still received radiation therapy within the 28-day benchmark; the change from 2019 was only about three percentage points.

This is evidence of resilience because service performance was comparatively preserved during a major system shock. It is also evidence of high restorability because performance remained close to the historical benchmark rather than settling into a substantially degraded post-shock state.

Shock → Service largely maintained → Benchmark performance remains near baseline

SRD should classify this as successful durable resilience, not compensated deterioration.

ADVERSARIAL TEST — Ceiling-effect criticism

Radiation therapy may have had stronger pre-existing capacity, protected prioritization, different workflow, or lower volume elasticity than other services.

The quadrant does not prove that resilience caused restorability. It only demonstrates that the two can jointly be high.

5. Quadrant II — Low Resilience, High Restorability:

Cataract Surgery

Cataract surgery provides the clearest successful-restoration control. CIHI reported that only 45% of patients received cataract surgery within the recommended 112 days in 2020, compared with substantially better pre-pandemic performance. By 2023, 70% were again treated within the benchmark, matching the pre-pandemic level; in 2024, performance remained approximately at baseline at 69% versus 70% in 2019.

Benchmark attainment: 2020 = 45% → 2023 = 70% → 2024 = 69%

The initial shock response therefore showed low resilience relative to radiation therapy: access deteriorated sharply. Yet the subsequent trajectory showed high restorability: the service returned close to its historical benchmark.

This quadrant is particularly important for SRD because it demonstrates that low resilience does not necessarily imply low restorability.

Low shock absorption ≠ inability to restore

ADVERSARIAL TEST — Alternative explanation

Cataract surgery can often be performed in day-procedure rooms or community clinics, unlike many joint replacements that depend on hospital operating rooms. CIHI has explicitly noted this difference. High restorability may therefore reflect service architecture rather than a general system property. That is acceptable: restorability is expected to depend on structure.

6. Quadrant III — Low Resilience, Low/Incomplete

Restorability: Joint Replacement

Joint replacement experienced substantial pandemic disruption and has recovered incompletely. CIHI reported that by 2024 hip-replacement volumes were 26% above 2019 and knee-replacement volumes 21% above 2019. Yet only 68% of hip replacements occurred within the 26-week benchmark compared with 75% in 2019, and only 61% of knee replacements met the benchmark compared with 70% in 2019.

Hip: Volume +26% vs 2019; benchmark 75% → 68%

Knee: Volume +21% vs 2019; benchmark 70% → 61%

This is exactly the sort of trajectory that motivated SRD. Output recovered and exceeded the historical volume, but requirement-adjusted access did not fully restore. CIHI notes that overall surgery volume growth has had to contend with population growth and a larger rise in surgical demand among older adults.

The resilience classification is low relative to radiation therapy because the service suffered major pandemic disruption and accumulated waits. Restorability is incomplete because benchmark attainment remained materially below 2019 in 2024 despite expanded output.

Output restoration ≠ access restoration

ADVERSARIAL TEST — Demand-denominator criticism

The benchmark gap may reflect higher demand rather than impaired restoration

capacity. SRD therefore cannot infer structural failure from the benchmark alone. The correct conclusion is incomplete restoration relative to the historical target under a higher requirement, which is precisely why SRD tracks SRR rather than raw output.

7. Quadrant IV — High Resilience, Low/Incomplete

Restorability: Hospital-Wide Continuity

The most consequential quadrant is High Resilience–Low Restorability. Canadian hospitals continued operating through the pandemic and post-pandemic period by adapting staffing and service arrangements. In 2023–2024, more than 7.8 million private-agency staffing hours were used, 126% above 2019–2020, and almost 32 million overtime hours were worked, 100% above 2019–2020. CIHI explicitly describes these mechanisms as ways hospitals fill urgent staffing gaps and maintain uninterrupted care. That is evidence of resilience: the system mobilized alternate capacity rather than ceasing operation.

But system-wide restoration remains incomplete. CIHI's 2026 wait-time reporting says emergency-department waits increased between 2018–2019 and 2024–2025; cancer-surgery and diagnostic-imaging delays persisted even while several priority services recovered. The staffing system itself also remained materially dependent on extraordinary hours.

High continuity + high adaptive response + persistent access/staffing burden

This is the candidate quadrant that matters most to SRD:

High Resilience $\wedge$ Low/Incomplete Restorability

The evidence is sufficient to identify the descriptive state, but not to prove that resilience caused the restoration deficit. Demand growth, aging, case complexity, workforce preferences, provincial heterogeneity, and pandemic backlog remain important alternative explanations.

ADVERSARIAL TEST — Strongest attack on SRD

Resilience engineering could simply call this 'resilience under continuing adverse

conditions' and include restoration as one dimension of adaptive capacity. If so, SRD has identified a useful sub-dimension but not necessarily an independent field-level variable. Only prospective incremental prediction can decide that.

8. The Matrix Does Not Collapse

If resilience and restorability were empirically identical in this test, the services should roughly align along one diagonal:

High Resilience → High Restorability

Low Resilience → Low Restorability

But the observed candidate trajectories include off-diagonal cases:

Cataract: Low Resilience → High Restorability

Hospital-wide continuity: High Resilience → Low/Incomplete Restorability

This establishes descriptive separability. It does not yet establish statistical independence, causal independence, or incremental predictive value.

ADVERSARIAL TEST — Post-hoc classification

The quadrants were selected after observing known trajectories. That means this paper is a construct test, not an out-of-sample validation. The taxonomy must now be frozen and applied prospectively to new services or periods.

9. Formal Distinction

The comparison suggests two different time-indexed properties.

$R_t = f(\text{continuity, adaptive response, disturbance absorption})$

$\mathcal{R}_{t,k} = f(\text{deficit closure, slack rebuilding, compensation sustainability, restoration lead time})$

A system can therefore occupy four states:

- R high, $\mathcal{R}$ high: it absorbs/adapts and subsequently restores durable capacity.
- R high, $\mathcal{R}$ low: it maintains function but restoration margin remains impaired.
- R low, $\mathcal{R}$ high: it suffers visible disruption but rebuilds effectively afterward.

- R low, $\mathcal{R}$ low: it neither absorbs disturbance well nor restores effectively.

The first two are especially important because conventional continuity metrics can look favorable in both while future restoration implications differ.

10. What 'Restorability' Must Not Mean

Restorability cannot simply be defined as 'resilience later.' That would make the distinction tautological. It must use variables not contained in the initial resilience classification.

For this research program, restorability requires evidence on:

- Requirement-adjusted deficit closure rather than raw activity alone.
- Rebuilding of durable slack or reserve.
- Declining dependence on depleting/extraordinary compensation where such compensation was used.
- Restoration lead time relative to remaining buffer or service need.
- Absence of equivalent burden transfer to another subsystem or population.

$$\text{Durable restorability evidence: } d\text{CRD}/dt < 0 \wedge d\text{Slack}/dt > 0 \wedge d\text{C_depleting}/dt \leq 0$$

The equation defines an evidentiary standard, not a currently calibrated universal score.

11. Resilience Can Contain Restorability

A serious alternative remains: perhaps mature resilience engineering already contains everything SRD calls restorability. Hollnagel, Woods, and later resilience work explicitly treat anticipation, adaptation, learning, and sustained adaptive capacity as central. NIST functional-recovery engineering also directly addresses target recovery times, organizational action, and system capacity.

SRD therefore does not claim that other fields are blind to restoration. Its narrower claim is that requirement-adjusted restorability should be measured as an explicit outcome even when observed continuity remains high.

The empirical contest is not terminology; it is predictive sufficiency.

If R-variables predict $\mathcal{R}$ outcomes fully, separate SRD restorability adds no value

ADVERSARIAL TEST — Parsimony test

If a resilience model using anticipation, monitoring, response, learning, reserve capacity, and adaptation predicts future deficit closure as well as an SRD model, the simpler or established model should be preferred.

12. Prospective Falsification Test

The 2×2 matrix is now frozen for prospective use. Future data should classify services before knowing the subsequent restoration outcome.

For each service i at time t :

$R_{i,t}$ = pre-specified resilience score/class

$\mathcal{R}_{i,t \rightarrow t+1}$ = later restorability outcome

Then estimate:

$M_R: \mathcal{R}_{\{t+1\}} = f(R_t, \text{demand, baseline capacity, trend})$

$M_{SRD}: \mathcal{R}_{\{t+1\}} = M_R + RD_t + CRD_t + C_{\text{depleting},t} + \text{Slack}_t + \tau_{R,t}$

SRD survives the incremental-value test only if M_{SRD} improves held-out or prospective prediction enough to justify the added variables.

$\text{Performance}(M_{SRD}) > \text{Performance}(M_R) \rightarrow \text{incremental SRD evidence}$

$\text{Performance}(M_{SRD}) \leq \text{Performance}(M_R) \rightarrow \text{restorability should be treated as redundant/nested}$

No post-hoc quadrant creation is allowed after the validation period begins.

13. A Stronger Negative Control

Cataract surgery now functions as a successful-restoration negative control against SRD overdiagnosis. It experienced a major shock-associated deterioration, yet returned close to 2019 benchmark performance. SRD must classify that trajectory as restored rather than continue warning simply because the broader healthcare system remains stressed.

Shock + deterioration + recovery to baseline = successful restorability

Radiation therapy supplies an additional control: high resilience need not conceal hidden deterioration when service performance remains stable and near benchmark.

These controls make SRD harder to rescue through selective case choice.

14. A Stronger Positive Test

Joint replacement provides a more demanding SRD test than hospital-wide rhetoric because it shows a concrete divergence between activity and access:

Volume ↑↑ while benchmark attainment remains below baseline

If future years show benchmark restoration while volumes stabilize and extraordinary system compensation declines, SRD should classify the restoration gap as closing. If benchmark performance remains impaired despite sustained high activity and adjusted demand controls, the restoration-deficit interpretation strengthens.

15. Cross-Domain External Validity

The healthcare matrix is intentionally the primary test because it minimizes cross-domain confounding. SRD's broader field claim, however, requires external validity. Electricity provides a candidate High Resilience–Prospective Low Restorability condition: current grid reliability can remain high while NERC forecasts 224 GW of additional summer peak demand over the next decade and warns that resource-addition lag may not keep pace with demand, much of it from AI and digital-economy data centres.

This is not used to prove the matrix. It generates a prospective prediction: a system can remain highly reliable while its future restoration/adequacy margin compresses.

Current Reliability High $\wedge$ Future Restoration Margin ↓

If dependable capacity, transmission, storage, and flexibility outpace realized demand growth, SRD should downgrade the compression diagnosis.

16. What Was Found

The attempt to break the resilience–restorability distinction produces a mixed but meaningful result.

- Descriptive separability survives: Canadian healthcare contains plausible off-diagonal cases.
- Cataract surgery demonstrates that low resilience during a shock can coexist with high subsequent restorability.
- Hospital-wide continuity demonstrates that high resilience can coexist with incomplete restoration indicators.
- Radiation therapy demonstrates that high resilience can coexist with high restorability; SRD therefore cannot diagnose hidden deficit from continuity alone.
- Joint replacement demonstrates that restoration of output can coexist with incomplete restoration of access.
- The result does not establish statistical independence, causal independence, or predictive superiority of SRD.

**Finding: Resilience and Restorability are descriptively separable;
incremental predictive value remains unproven.**

17. Implications for SRD

This test narrows SRD further. Restorability should be treated as the field's core dependent variable, not as an automatic consequence of deficit language.

The field's strongest surviving question becomes:

**After adaptation preserves or restores service, has the system rebuilt
durable capacity relative to its changing structural requirement?**

This formulation is compatible with resilience engineering and functional recovery while retaining a specific empirical burden.

SRD therefore shifts from a theory of hidden failure toward a theory of restoration sufficiency.

18. Limitations

- The quadrant assignments are based on aggregate national indicators and are not formal latent-variable estimates.
- Resilience is multidimensional; service continuity during COVID is only a partial proxy.

- 2019 is a historical baseline, not a normative optimum.
- Canadian healthcare services differ in workflow, location, resource requirements, urgency, and treatment setting.
- Hospital-wide continuity is broader than procedure-specific restorability and therefore not perfectly comparable.
- Compensation sustainability is not directly estimated in this paper.
- Demand adjustment is discussed but not fully modeled quantitatively.
- The 2×2 matrix is derived retrospectively and requires prospective validation.
- The electricity example is forecast-based and is included only for external-validity hypothesis generation.

18.1 Establishment Pass: What Can Be Established Now

The preceding sections identified three unresolved propositions. This establishment pass separates what can be established from public evidence now from what requires new primary data. The standard is intentionally asymmetric: absence of the required measurement is not treated as support for SRD.

18.2 Claim 1 — Statistical Distinctness of Resilience and Restorability

Verdict: NOT ESTABLISHED with the present Canadian administrative dataset.

Reason: a proper resilience-engineering comparator requires independent measurement of resilience potentials rather than inference from service continuity. The ED-RAG literature operationalizes resilience through respond, monitor, learn, and anticipate. The validated emergency-department instrument contains 38 questions across 32 foci and is designed as a context-specific snapshot that can be repeated over time. CIHI administrative data provide staffing, vacancy, overtime, sick-leave, supply, and service indicators, but do not supply a national RAG-equivalent measure.

Therefore the current off-diagonal matrix establishes outcome separability only:

Performance during disturbance $\neq$ subsequent restoration trajectory

It does not establish:

Latent resilience construct R statistically independent of restorability $\mathcal{R}$

Required evidence: repeated hospital-level RAG/ED-RAG measurements linked to independently constructed SRD restoration outcomes. Factor/discriminant validity and longitudinal correlation can then test whether the constructs collapse.

18.3 Claim 2 — Incremental Predictive Value of Restorability

Verdict: NOT ESTABLISHED; public data are sufficient to build the SRD side of the test, but not a matched national resilience-potential comparator.

CIHI's updated health-workforce resources contain overtime, sick-leave and vacancy indicators, and its wait-time series provide future service outcomes. These data make a longitudinal restoration model feasible. However, replacing RAG resilience with continuity proxies would prejudge the comparison in SRD's favor.

The admissible head-to-head test remains:

$$\mathbf{M_R: Y_}(t+1) = \mathbf{f(RAG_t, demand_t, baseline\ capacity_t, trend)}$$

$$\mathbf{M_SRD: Y_}(t+1) = \mathbf{M_R + RD_t + CRD_t + C_depleting,t + Slack_t + \tau_R,t}$$

The test must use outcomes not used to construct either predictor set and must be evaluated out of sample. Until matched RAG and restoration data exist, the paper cannot claim SRD predicts more.

18.4 Claim 3 — Does Persistent Compensation Damage Future Capacity?

Verdict: PARTIALLY SUPPORTED AS A HUMAN-SUBSTRATE MECHANISM; NOT ESTABLISHED AS THE CANADIAN SYSTEM-LEVEL SRD CAUSAL LOOP.

External longitudinal evidence strengthens the plausibility of the causal mechanism. A four-year healthcare-worker study found prolonged overtime predicted worsening burnout over time. Separate hospital evidence has associated mandatory overtime with intent to leave. These findings make a pathway from sustained extraordinary work to human-capacity pressure empirically credible.

But they do not establish the complete SRD system loop:

$$\mathbf{StaffingDeficit \rightarrow Compensation \rightarrow Attrition/CapacityLoss \rightarrow StaffingDeficit}$$

for Canadian hospitals. The available Canadian aggregate data establish that overtime and purchased staffing are used to fill staffing gaps; they do not identify the counterfactual effect of those mechanisms on later staffing capacity.

Required evidence: a hospital/province panel linking lagged overtime and agency dependence to subsequent exits, vacancies, absence, retention, access, and staffing, controlling for initial shortage, demand, wages, policy, case mix, and regional labor supply.

18.5 Strongest Result That Is Established

The strongest current empirical result is narrower than the original SRD field claim but robust to the adversarial review:

A system's performance during disturbance and its subsequent requirement-adjusted restoration trajectory are different empirical outcomes.

Canadian procedure trajectories demonstrate this directly. Cataract access deteriorated sharply and later returned close to its 2019 benchmark, while joint-replacement volumes exceeded 2019 levels without equivalent restoration of benchmark attainment. This establishes that shock absorption, activity recovery, and access restoration cannot be treated as a single variable.

It does not, by itself, prove that restorability is independent of a fully measured resilience construct.

18.6 Status Table After Establishment Attempt

Claim	Status	Reason
Performance during disturbance differs from later restoration	ESTABLISHED	Observed off-diagonal Canadian service trajectories.
Raw output recovery differs from access restoration	ESTABLISHED	Joint-replacement volume and benchmark attainment diverge.
Resilience construct is statistically distinct from restorability	NOT ESTABLISHED	No matched independent RAG/ED-RAG and SRD measurements.

SRD restorability predicts future outcomes beyond resilience	NOT ESTABLISHED	Head-to-head prospective model cannot yet be run fairly.
Persistent overtime can pressure the human substrate	SUPPORTED OUTSIDE CANADA	Longitudinal overtime–burnout evidence; mandatory overtime–intent-to-leave evidence.
Compensation causes future Canadian system capacity loss	NOT ESTABLISHED	Aggregate Canadian data do not identify the causal effect.
SRD is irreducible to resilience engineering	UNRESOLVED	Depends on construct and incremental-prediction tests.

18.7 Pre-Registered Next Validation Design

The next empirical study should recruit multiple emergency departments or hospitals, administer the same tailored resilience instrument at baseline, construct SRD restoration measures independently, and freeze predictions before the next observation wave.

Primary falsifier:

If RAG_t explains future restoration outcomes and {RD, CRD, C_{depleting}, Slack, τ_R } add no reliable held-out information, restorability should be nested inside resilience rather than claimed as an independent SRD contribution.

Primary confirmation threshold:

SRD earns incremental evidence only if independently measured restoration variables improve prospective prediction or intervention classification beyond resilience potentials, with the result replicated across sites.

19. Conclusion

The surviving SRD proposition withstands an initial attempt at empirical destruction, but only in a limited form. Resilience and restorability do not map perfectly onto one another in the Canadian healthcare trajectories examined. Some services were disrupted sharply and later restored; others maintained high continuity yet still exhibit unresolved system-level restoration pressures.

Resilience $\neq$ automatically Restorability

That statement is now descriptively supported, but it is not enough to establish an independent scientific contribution. The decisive question is prospective: do SRD's requirement-adjusted restoration variables predict future deficit closure, slack rebuilding, or restoration difficulty beyond established resilience measures?

If the answer is no, restorability should be treated as a nested resilience dimension. If the answer is yes, SRD has identified an empirically distinct property of complex systems.

SRD survives only if restorability adds information.

That is the next test.

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